SHOPIFY SALES & CUSTOMER ANALYTICS

AI-assisted dashboard project | March 18–24, 2025 | USD

Project objective
Analyse the observed Shopify transaction data to understand sales performance, product contribution, geographic distribution, payment gateway mix, and customer purchasing frequency.

Prepared from the cleaned dataset and the Julius AI dashboard screenshots provided. Metrics below reflect the dashboard as shown and should be validated against the source workbook before formal submission.

1. Executive Summary

The Julius AI dashboard summarises 7,431 observed orders from March 18–24, 2025. It reports total recorded sales of $4,598,960.87, an average order value of $618.89, and 4,431 distinct observed customer IDs. Running Shoes is the leading product type by recorded sales, while California is the top billing state shown.

Metric	Dashboard value	Interpretation
Total recorded sales	$4,598,960.87	Tax-inclusive recorded sales, per dashboard label
Total orders	7,431	Distinct order numbers as reported
Average order value	$618.89	Recorded sales divided by orders
Unique customers	4,431	Distinct observed customer IDs

2. Dataset and Scope

The dashboard identifies the source as the cleaned_sales worksheet and shows a seven-day reporting window from March 18 through March 24, 2025. The observed data is labelled USD and uses United States billing locations. The dashboard reports 7,431 rows and 17 columns in the cleaned worksheet.

The analysis is descriptive and limited to the supplied observation window. It should not be interpreted as a full-year trend or a reliable measure of long-term customer retention or lifetime value.

3. Data Preparation and Quality

The portfolio copy was prepared by standardising column names, trimming whitespace in text fields, normalising selected text values, parsing invoice dates as date/time, and converting numeric fields to numeric types where possible. Direct customer first and last name columns were removed from the portfolio copy; the original source file was not changed.

The data-quality check found no exact duplicate rows. The dashboard and cleaning log identify 11 missing Product IDs and 4 missing Variant IDs. These records were retained rather than deleting potentially useful sales rows or guessing missing identifiers.

4. Dashboard Components

- KPI cards for total recorded sales, total orders, average order value, and unique customers.
- Daily sales trend for March 18–24, 2025.
- Revenue by product type across 14 categories.
- Revenue by billing state/district and city/state combinations.
- Revenue by payment gateway across five gateways.
- Customer purchase-frequency distribution and observed customer-segment sales.
- Dataset inspection and quality-audit section.

5. Key Observations from the Dashboard

- **Running Shoes is the largest product category:** $1,616,946.28, or 35.2% of recorded sales.
- **California leads the displayed billing states:** reported sales are $535,786.61.
- **Shopify Payments is the largest recorded payment gateway:** it accounts for 58.5% of recorded sales. The dashboard notes that gateway labels do not establish payment settlement.
- **Repeat orders are common within the observed week:** 2,039 customers (46.0%) placed two or more observed orders and contributed 67.8% of sales.
- **March 24 is the peak observed day:** $752,227.32 across 1,204 orders.

These findings describe the seven-day dataset only. “Repeat” means at least two distinct observed orders during the window; it does not prove long-term customer loyalty.

6. Business Recommendations

- Review inventory and availability for Running Shoes, while monitoring margin and stock-out rates before increasing purchasing commitments.
- Investigate what drives sales in California and compare customer mix, order volume, and average order value across other states.
- Use payment gateway mix to prioritise checkout reliability monitoring; validate settlement and refund records separately.
- Study customers with multiple observed orders to understand product combinations and purchasing patterns, without labelling them long-term loyal customers based on one week.
- Continue tracking sales over a longer period before drawing conclusions about seasonality, retention, or lifetime value.

7. Limitations and Validation Notes

- The reporting window covers only seven days, so long-term retention, cohort analysis, seasonality, and customer lifetime value cannot be established reliably.
- The figures in this report are transcribed from the dashboard screenshots; validate them against the workbook and Julius calculations before formal submission.
- The dashboard labels the KPI “Total Recorded Sales” and says it includes tax. Do not automatically describe it as net revenue or profit.
- The customer repeat-order metric is limited to the observed window and should not be confused with first-ever versus returning customer status.
- The screenshots show a Julius AI dashboard; they do not demonstrate a Power BI .pbix file or DAX implementation.

8. Conclusion

This project demonstrates how transaction data can be transformed into a business dashboard using AI-assisted analysis. The dashboard highlights category concentration, geographic differences, payment gateway mix, and purchasing frequency. The next improvement is to validate each metric against the source data and, if required by the project brief, recreate the report in Power BI with documented DAX measures and interactive filters.

Portfolio note: Share only the anonymised cleaned workbook and screenshots that do not expose customer personal information. Keep this report alongside the dashboard preview and README.